Report

Subject Mathematical Logic

Submitted to

Assistant Professor Dr.Iitiwat Tocharoennirattisa

Prepared by

Ms. Asma Cheloh ID 6711070055

Ms. Nur-aima mahae ID 671107068

This report is part of the English for teaching mathematics

Semester 1 Academic Year 2026

Faculty of Education Mathematics Department

University Rajabhat Thonbur

Preface

This report was prepared to learn and better understand Mathematical Logic, one of the fundamental areas of mathematics that focuses on logical thinking and correct reasoning. The topics covered in this report include propositions, logical connectives, truth tables, logical equivalence, quantifiers, rules of inference, and methods of mathematical proof. These concepts are essential because they help develop logical thinking, problem-solving skills, and the ability to construct valid mathematical arguments. They also serve as an important foundation for further studies in mathematics, computer science, and related fields.

While preparing this report, the author studied information from textbooks and other reliable academic sources to gain a clearer understanding of the subject. It is hoped that this report will not only help the author strengthen their knowledge of mathematical logic but also be useful for anyone who is interested in this topic. The author sincerely appreciates the guidance and support provided by the instructor throughout this work and welcomes any comments or suggestions that may help improve future studies.

Asma Cheloh

Nur-aima mahae

Table of contents

Chapter 1 Introduction to Mathematical Logic

Chapter 2 Logical Connectives and Truth Tables

Chapter 3 Tautologies

Chapter 4 Arguments

Chapter 5 Open Sentences and Quantifiers

Chapter 6 Inductive and Deductive Reasoning

Conclusion / References

Page numbering can be updated after completing the cover.

Notation and Proof Conventions

Symbol	Meaning
p, q, r, s, t	Atomic propositions
¬	Negation
∧	Conjunction
∨	Inclusive disjunction
→	Material implication
↔	Biconditional
⊢	Syntactic derivability
⊨	Semantic entailment
∀, ∃	Universal and existential quantifiers

Chapter 1. Introduction to Mathematical Logic

Definitions and central ideas

Mathematical logic studies the formal structure of valid reasoning. A proposition has exactly one truth value, true or false. Atomic propositions are represented by p, q, r, and s; compound propositions are formed using connectives. Logical validity concerns whether a conclusion must be true whenever all premises are true, not whether the premises happen to describe actual events.

Theory and proof principles

Syntax specifies which strings are well-formed formulas; semantics assigns truth values to formulas under a valuation. A formula A is satisfiable if at least one valuation makes A true; it is valid if every valuation makes A true. A set Γ entails B ($\Gamma \models B$) if no valuation makes every formula in Γ true and B false. A derivation $\Gamma \vdash B$ is a sequence justified by proof rules.

Advanced worked proofs

Proof 1.1: Chained inference with four propositions

Premises: $p \longrightarrow q$; $q \longrightarrow r$; $r \longrightarrow s$; p

Prove: s

Proof:

1. From p and $p \longrightarrow q$ infer q (modus ponens).

2. From q and $q \longrightarrow r$ infer r (modus ponens).

3. From r and $r \longrightarrow s$ infer s (modus ponens).

Therefore the conclusion follows from the premises.

Proof 1.2: Indirect proof using three atoms

Premises: $p \vee q$; $p \longrightarrow r$; $q \longrightarrow r$

Prove: r

Proof:

1. Assume $\neg r$ for contradiction.

2. By modus tollens on $p \longrightarrow r$ and $\neg r$, infer $\neg p$.

3. By modus tollens on $q \longrightarrow r$ and $\neg r$, infer $\neg q$.

4. From $p \vee q$ and $\neg p$ infer q; this contradicts $\neg q$.

5. Discharge $\neg r$ and conclude r.

Therefore the conclusion follows from the premises.

.

Chapter 2. Logical Connectives and Truth Tables

Definitions and central ideas

Negation $\neg p$ reverses truth. Conjunction $p \wedge q$ is true only when both conjuncts are true. Inclusive disjunction $p \vee q$ is false only when both are false. Implication $p \rightarrow q$ is false exactly when p is true and q is false. Biconditional $p \leftrightarrow q$ is true when p and q have the same truth value. The symbol $\rightarrow$ expresses material implication, not necessarily a causal relationship.

Theory and proof principles

A truth table for n distinct atomic propositions contains 2^n rows. Evaluate subformulas from the innermost parentheses outward. Two formulas are logically equivalent exactly when their final columns match in every row. Useful identities include

$$A \rightarrow B \equiv \neg A \vee B, \neg(A \wedge B) \equiv \neg A \vee \neg B, and \neg(A \vee B) \equiv \neg A \wedge \neg B.$$

Three-variable truth table

p	q	r	$(p \wedge q) \rightarrow r$	$\neg p \vee \neg q \vee r$	$\leftrightarrow$
T	T	T	T	T	T
T	T	F	F	F	T
T	F	T	T	T	T
T	F	F	T	T	T
F	T	T	T	T	T
F	T	F	T	T	T
F	F	T	T	T	T
F	F	F	T	T	T

The final column is T in all eight rows; therefore the two formulas are equivalent.

Four-variable truth table

p	q	r	s	$(p \wedge q) \rightarrow (r \vee s)$	$\neg p \vee \neg q \vee r \vee s$	$\leftrightarrow$
T	T	T	T	T	T	T
T	T	T	F	T	T	T
T	T	F	T	T	T	T
T	T	F	F	F	F	T
T	F	T	T	T	T	T
T	F	T	F	T	T	T
T	F	F	T	T	T	T
T	F	F	F	T	T	T
F	T	T	T	T	T	T
F	T	T	F	T	T	T
F	T	F	T	T	T	T
F	T	F	F	T	T	T
F	F	T	T	T	T	T
F	F	T	F	T	T	T
F	F	F	T	T	T	T
F	F	F	F	T	T	T

Advanced worked proofs

Proof 2.1: Distribution of implication over conjunction

Premises: $p \rightarrow (q \wedge r)$

Prove: $(p \rightarrow q) \wedge (p \rightarrow r)$

Proof:

1. Rewrite implication as $\neg p \vee (q \wedge r)$.

2. Distribute $\vee$ over $\wedge$: $(\neg p \vee q) \wedge (\neg p \vee r)$.

3. Rewrite both disjunctions as implications.

4. Therefore the formulas are equivalent.

Therefore the conclusion follows from the premises.

Proof 2.2: Disjunction in an antecedent

Premises: $(p \vee q) \rightarrow r$

Prove: $(p \rightarrow r) \wedge (q \rightarrow r)$

Proof:

1. Rewrite as $\neg(p \vee q) \vee r$.

2. Apply De Morgan: $(\neg p \wedge \neg q) \vee r$.

3. Distribute: $(\neg p \vee r) \wedge (\neg q \vee r)$.

4. Rewrite as $(p \rightarrow r) \wedge (q \rightarrow r)$.

Therefore the conclusion follows from the premises.

Proof 2.3: A four-atom equivalence

Premises: $(p \wedge q) \rightarrow (r \vee s)$

Prove: $\neg p \vee \neg q \vee r \vee s$

Proof:

1. Eliminate implication: $\neg(p \wedge q) \vee (r \vee s)$.

2. Apply De Morgan: $(\neg p \vee \neg q) \vee (r \vee s)$.

3. Use associativity to obtain $\neg p \vee \neg q \vee r \vee s$.

Therefore the conclusion follows from the premises.

Example1 : Determine the truth value of $[(q \rightarrow s) \vee r] \vee [(q \leftrightarrow s) \wedge r]$

Proof : Assume the proposition is false

$$[(q \rightarrow s) \vee r] \equiv F$$
$$q \rightarrow s \equiv F \Longrightarrow q \equiv T, s \equiv F$$
$$r \equiv F$$

Substitute

$$q \equiv T, s \equiv F, r \equiv F :$$

$$[(q \leftrightarrow s) \wedge r] \equiv F$$

Chapter 3. Tautologies

Definitions and central ideas

A tautology is true under every truth assignment; a contradiction is false under every assignment; a contingent formula is true under some assignments and false under others. To establish that A is a tautology, one may construct its full truth table or derive it from valid inference and equivalence rules. To disprove tautologicity, one countervaluation suffices.

Theory and proof principles

The deduction theorem connects valid arguments to tautologies: premises $A_1, \ldots, A_n$ entail C exactly when $(A_1 \wedge \cdots \wedge A_n) \rightarrow C$ is valid in classical propositional logic. In natural deduction, temporary assumptions must be discharged correctly; an unlicensed step can invalidate an otherwise convincing proof.

How to test a proposed tautology

For a formula with $p, q, r, s,$ check all sixteen valuations, or assume its negation and derive a contradiction. If a countervaluation exists, the formula is not a tautology.

Advanced worked proofs

Proof 3.1: Transitivity tautology

Premises: $(p \rightarrow q) \wedge (q \rightarrow r)$

Prove: $p \rightarrow r$

Proof:

1. Assume both implications.

2. Assume p temporarily.

3. From $p \rightarrow q$ and p infer q; from $q \rightarrow r$ and q infer r.

4. Discharge p to infer $p \rightarrow r.$

5. Discharge the initial conjunction to establish the tautology.

Therefore the conclusion follows from the premises.

Proof 3.2: Constructive dilemma tautology

Premises: (p ⟶ r) ⋀ (q ⟶ r) ⋀ (p ⋁ q)

Prove: r

Proof:

1. Extract $p \rightarrow r, q \rightarrow r, and\ p \vee q.$

2. In the p case, infer r using $p \rightarrow r.$

3. In the q case, infer r using $q \rightarrow r.$

4. Apply disjunction elimination to conclude r.

5. Thus the corresponding implication is a tautology.

Therefore the conclusion follows from the premises.

Proof 3.3: Joint contrapositive tautology

Premises: $(p \rightarrow q) \wedge (r \rightarrow q) \wedge \neg q$

Prove: $\neg p \wedge \neg r$

Proof:

1. Extract $p \rightarrow q, r \rightarrow q, and \neg q.$

2. From $p \rightarrow q\ and \neg q\ infer \neg p.$

3. From $r \rightarrow q\ and \neg q\ infer \neg r.$

4. Conjoin the results and discharge the assumptions.

Therefore the conclusion follows from the premises.

Proof 3.4: Four-variable elimination tautology

Premises: $p \lor q \lor r \lor s;\ \neg p;\ \neg q;\ \neg r$

Prove: s

Proof:

1. Eliminate p from $p \lor (q \lor r \lor s)\ using\ \neg p.$

2. Eliminate q from $q \lor r \lor s\ using\ \neg q.$

3. Eliminate r from $r \lor s\ using\ \neg r.$

4. Conclude s; hence the implication from the conjunction of premises is valid.

Therefore the conclusion follows from the premises.

Chapter review

The examples demonstrate that complex formulas should be decomposed into smaller statements and that each inference must be justified. The distinction between validity, truth, and equivalence remains essential.

example1 :Is the following proposition a tautology?

$$(p \wedge q) \rightarrow [(p \vee q) \rightarrow r]$$

Proof

$$(p \wedge q) \rightarrow [(p \vee q) \rightarrow r] \equiv F$$

$$p \wedge q \equiv T \Longrightarrow p \equiv T, q \equiv T$$
$$(p \vee q) \rightarrow r \equiv F \Longrightarrow (p \vee q) \equiv T, r \equiv F$$

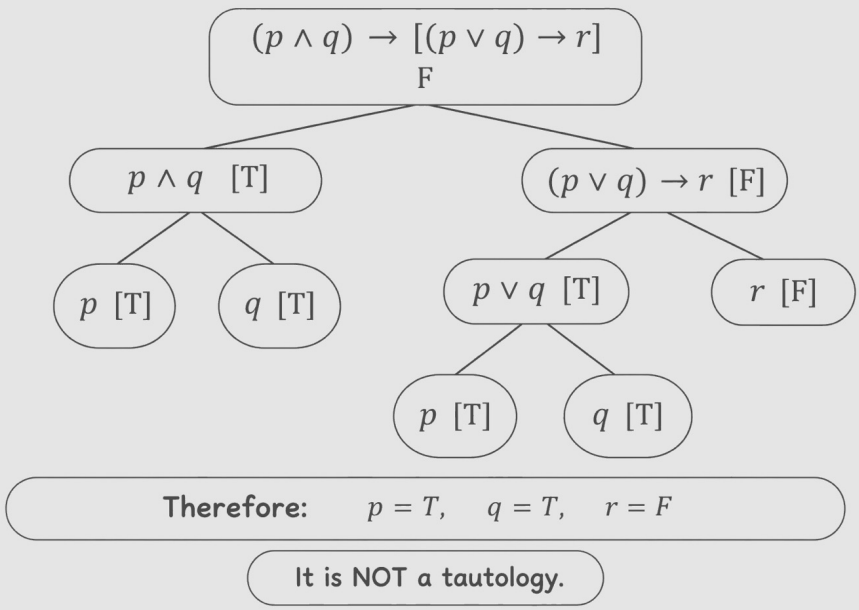
$$p \equiv T, q \equiv T \quad p \vee q \quad T \vee T \equiv T$$
$$p \equiv T, q \equiv T, r \equiv F$$

It is Not a tautology.

Example 1

Is the following proposition a tautology?

$(p \wedge q) \rightarrow [(p \vee q) \rightarrow r]$



example2 :Is the following proposition a tautology?

$$[(p \wedge q) \rightarrow (p \vee q)] \leftrightarrow [(\sim p \wedge \sim q) \rightarrow (\sim p \vee \sim q)]$$

Proof $[(p \wedge q) \rightarrow (p \vee q)] \leftrightarrow [(\sim p \wedge \sim q) \rightarrow (\sim p \vee \sim q)]$

$$(p \wedge q) \equiv T \Longrightarrow p \equiv T, q \equiv T$$
$$(p \vee q) \equiv T \Longrightarrow (p \wedge q) \rightarrow (p \vee q) \equiv T$$

$$(\sim p \wedge \sim q) \equiv T \Rightarrow \sim p \equiv T, \sim q \equiv T$$
$$(\sim p \vee \sim q) \equiv T \Rightarrow (\sim p \wedge \sim q) \rightarrow (\sim p \vee \sim q) \equiv T$$

$$T \leftrightarrow T \equiv T$$

It is a tautology.

example3 : Pove that $[(\sim p \wedge q) \rightarrow (\sim q \wedge r)] \rightarrow (\sim r \wedge s)$ is a tautology

Proof Assume the proposition is fals

$$[(\sim p \wedge q) \rightarrow (\sim q \wedge r)] \rightarrow (\sim r \wedge s) \equiv F$$

$$[(\sim p \wedge q) \rightarrow (\sim q \wedge r)] \equiv T$$

$$(\sim r \wedge s) \equiv F$$

Assume $(\sim p \wedge q) \equiv T \Longrightarrow p \equiv F, q \equiv T$

$$\sim q \wedge r \equiv F \wedge r \equiv F$$

$[(\sim p \wedge q) \rightarrow (\sim q \wedge r)] \equiv T \Longrightarrow F \equiv F$ (Contradiction with T)

∴The proposition is a Tautology (always True)

Example4 : Is the following proposition a tautology ?

$$[(p \wedge q) \rightarrow r] \leftrightarrow [p \rightarrow (q \rightarrow r)]$$

Proof

$$[(p \wedge q) \rightarrow r] \leftrightarrow [p \rightarrow (q \rightarrow r)] \equiv F$$

$$[(p \wedge q) \rightarrow r] \equiv T \; and \; [p \rightarrow (q \rightarrow r)] \equiv F$$

or

$$[(p \wedge q) \rightarrow r] \equiv F \; and \; [p \rightarrow (q \rightarrow r)] \equiv T$$

Consider the case 1 of $[p \rightarrow (q \rightarrow r)] \equiv F$

$$p \equiv T \; and \; (q \rightarrow r) \equiv F$$

$$(q \rightarrow r) \equiv F \Longrightarrow q \equiv T, r \equiv F$$

Bring the value $p \equiv T, q \equiv T, r \equiv F$ Go instead on the shore $[(p \wedge q) \rightarrow r]$:

$$(p \wedge q) \equiv (T \wedge T) \equiv T$$

$$(p \wedge q) \rightarrow r \equiv T \rightarrow F \equiv F$$

There is a conflict with the original conditions set by $[(p \wedge q) \rightarrow r] \equiv T$

Consider the case 2 of $[(p \wedge q) \rightarrow r] \equiv F$:

$$(p \wedge q) \equiv T \; and \; r \equiv F$$
$$(p \wedge q) \equiv T \Longrightarrow p \equiv T, q \equiv T$$

Bring the value $p \equiv T, q \equiv T, r \equiv F$ Go instead on the shore $[p \rightarrow (q \rightarrow r)]$:

$$(q \rightarrow r) \equiv (T \rightarrow F) \equiv F$$
$$p \rightarrow (q \rightarrow r) \equiv T \rightarrow F \equiv F$$

There is a conflict with the original conditions set by $[p \rightarrow (q \rightarrow r)] \equiv T$

Summary Both cas1 and cas2 have aconflict.Therefoe,the assumption that

$[(p \wedge q) \rightarrow r] \leftrightarrow [p \rightarrow (q \rightarrow r)] \neq F$

As a result $[(p \wedge q) \rightarrow r] \leftrightarrow [p \rightarrow (q \rightarrow r)] \equiv T$ Be Tautology

Chapter 4. Arguments

Definitions and central ideas

An argument consists of premises and a conclusion. An argument is valid if every interpretation satisfying the premises also satisfies the conclusion. Soundness is stronger: an argument is sound when it is valid and its premises are true. Validity can be checked by natural deduction, truth tables, or a countermodel.

Theory and proof principles

Key valid patterns are modus ponens $(A \to B, A \vdash B)$, modus tollens $(A \to B, \neg B \vdash \neg A)$, hypothetical syllogism $(A \to B, B \to C \vdash A \to C)$, and disjunction elimination. Affirming the consequent $(A \to B, B \vdash A)$ and denying the antecedent $(A \to B, \neg A \vdash \neg B)$ are invalid.

A countermodel for an invalid inference

The pattern $p \to q, q, r \vee \neg r \vdash p$ is invalid. Choose $p \equiv F, q \equiv T, r \equiv F$: every premise is true while p is false. The extra third proposition does not repair the invalid inference.

Advanced worked proofs

Proof 4.1: Resolution with conditional branches

Premises: $p \vee q;\ p \to r;\ q \to s;\ \neg r$

Prove: s

Proof:

1. From $p \to r \ and \ \neg r \ infer \ \neg p.$

2. From $p \vee q \ and \ \neg p \ infer \ q.$

3. From $q \to s \ and \ q \ infer \ s.$

Therefore the conclusion follows from the premises.

Proof 4.2: Proof by contradiction with a compound consequent

Premises: $p \rightarrow (q \wedge r);\ \neg q \vee \neg r;\ p \vee s$

Prove: s

Proof:

1. By De Morgan, $\neg q \vee \neg r\ is\ \neg(q \wedge r).$

2. From $p \rightarrow (q \wedge r)\ and\ \neg(q \wedge r)\ infer\ \neg p.$

3. From $p \vee s\ and\ \neg p\ infer\ s.$

Therefore the conclusion follows from the premises.

Proof 4.3: Conditional proof with four atoms

Premises: $p \rightarrow q;\ (q \wedge r) \rightarrow s$

Prove**:** $(p \wedge r) \rightarrow s$

Proof:

1. Assume $p \wedge r;\ extract\ p\ and\ r.$

2. Use $p \rightarrow q\ to\ infer\ q.$

3. Combine q and r to obtain $q \wedge r$.

4. Apply $(q \wedge r) \rightarrow s$ to infer s.

5. Discharge $p \wedge r$ and conclude the conditional.

Therefore the conclusion follows from the premises.

Proof 4.4: Four-premise derivation

Premises: $p \rightarrow q;\ q \rightarrow r;\ r \rightarrow s;\ p \wedge \neg t$

Prove: $s \wedge \neg t$

Proof:

1. Extract p and $\neg t$ from $p \wedge \neg t.$

2. Apply modus ponens successively to derive q, r, and s.

3. Conjoin s with $\neg t$ to obtain $s \wedge \neg t.$

Therefore the conclusion follows from the premises.

Chapter review

The examples demonstrate that complex formulas should be decomposed into smaller statements and that each inference must be justified. The distinction between validity, truth, and equivalence remains essential.

proof

1. $(p \rightarrow q) \wedge (r \rightarrow s$	Premise
2. $p \vee r$	Premise
3. $\rightarrow q \rightarrow t$	Premise
4. $\rightarrow s \rightarrow t$	Premise
5. $p \rightarrow q$	Simplification
6. $r \rightarrow s$	Simplification
7. $q \vee s$	Constructive Dilemma
8. $q \vee s \vee t$	Addition

proof

1. $(p \rightarrow q) \wedge (r \rightarrow s$	Premise
2. $p \vee r$	Premise
3. $\rightarrow q \rightarrow t$	Premise
4. $\rightarrow s \rightarrow t$	Premise
5. $p \rightarrow q$	Simplification
6. $r \rightarrow s$	Simplification
7. $q \vee s$	Constructive Dilemma
8. $q \vee s \vee t$	Addition

Chapter 5. Open Sentences and Quantifiers

Definitions and central ideas

An open sentence $P(x)$ has a free variable and does not become a closed proposition until a value is assigned or the variable is bound. The universal quantifier $\forall x\ P(x)$ means that P holds for every element of the domain; the existential quantifier $\exists x\ P(x)$ means that P holds for at least one element. The domain must be stated explicitly.

Theory and proof principles

Quantifier negation obeys $\neg\forall x\ P(x)\ \equiv\ \exists x\ \neg P(x)\ and\ \neg\exists x\ P(x)\ \equiv\ \forall x\ \neg P(x).$ Quantifier order matters: $\forall x\exists y\ R(x,y)$ generally differs from $\exists y\forall x\ R(x,y).$ Existential elimination introduces a fresh witness symbol; universal introduction requires a genuinely arbitrary element.

Order of quantifiers: counterexample

Over the integers, $\forall x\exists y(x\ <\ y)$ is true: choose $y\ =\ x\ +\ 1$. However, $\exists y\forall x(x\ <\ y)$ is false because no integer is greater than every integer.

Advanced worked proofs

Proof 5.1: Universal-to-existential chain

Premises: $\forall x(P(x)\ \rightarrow\ Q(x));\ \forall x(Q(x)\ \rightarrow\ R(x));\ \exists x\ P(x)$

Prove: $\exists x\ R(x)$

Proof:

1. Choose a fresh witness a with $P(a)$ from the existential premise.

2. Instantiate the universal premises to obtain $P(a)\ \rightarrow\ Q(a)\ and\ Q(a)\ \rightarrow\ R(a).$

3. Infer $Q(a)$ and then $R(a)$.

4. Introduce the existential quantifier: $\exists x\ R(x).$

5. Discharge the fresh-witness subproof.

Therefore the conclusion follows from the premises.

Proof 5.2: Universal implication composition

Premises: $\forall x(P(x) \rightarrow Q(x));\ \forall x(Q(x) \rightarrow R(x))$

Prove: $\forall x(P(x) \rightarrow R(x))$

Proof: 1. Let a be an arbitrary domain element.

2. Assume $P(a).$

3. Instantiate both universal premises at $a.$

4. Use modus ponens twice to derive $R(a).$

5. Discharge $P(a)$ and generalize over arbitrary $a.$

Therefore the conclusion follows from the premises.

Chapter review

The examples demonstrate that complex formulas should be decomposed into smaller statements and that each inference must be justified. The distinction between validity, truth, and equivalence remains essential.

example1 : Prove that

$$(\exists x)(P(x) \wedge Q(x)) \rightarrow (\exists x)P(x)$$

Proof

Assume

$$(\exists x)(P(x) \wedge Q(x)) \rightarrow (\exists x)P(x)$$

$$= P(a) \wedge Q(a)$$
$$= P(a)$$
$$= (\exists x)P(x)$$
$$= (\exists x)(P(x) \wedge Q(x)) \rightarrow (\exists x)P(x)$$

Example 2 : Valid Argument (Tautology)

Problem : Prove whether the following statement is true or false :

$$(\forall x)\big(P(x) \vee Q(x)\big) \wedge (\exists x)\neg P(x) \rightarrow (\exists x)Q(x)$$

Proof :

Assume :

$$(\forall x)(P(x) \vee Q(x)) \wedge (\exists x)\neg P(x) \rightarrow (\exists x)Q(x)$$

Step 1: = $(\exists x)\neg P(x)$ (Given premise)

Step 2: = $\neg P(a)$ (by Existential Instantiation for some element a)

Step 3: = $(\forall x)(P(x) \vee Q(x))$ (Given premise)

Step 4: = $P(a) \vee Q(a)$ (by Universal Instantiation for element a)

Step 5: = $Q(a)$ (by Disjunctive Syllogism since $\neg P(a)$ is true)

Step 6: = $(\exists x)Q(x)$ (by Existential Generalization)

Answer : The statement is True (It is a tautology).

Example 3 : Invalid Argument (Not a Tautology)

Problem : Prove whether the following statement is a tautology or not :

Proof (By Counterexample) :

- Define Universe of Discourse (u) : $Let = \{1,2\}$
- Define Open Sentences : $Let\ P(x)\ mean' x = 1' \text{and}' \text{x} = 2'$

Step 1 (Evaluate the Premise) :

$$(\forall x)\big(P(x) \vee Q(x)\big)$$
$$= (1 = 1 \vee = 2) \wedge (2 = 1 \vee 2 = 2)$$
$$= (True \vee False) \wedge (True \vee False) = True$$

Step 2 (Evaluate the Conclusion) :

$$\big((\forall x)P(x)\big) \vee \big((\forall x)Q(x)\big)$$
$$= False \vee False = False$$

Step 3 (Check the Implication) :

$$Premise \rightarrow Conclusion$$

Answer : The statement is False (It is not a tautology).

Chapter 6. Inductive and Deductive Reasoning

Definitions and central ideas

Inductive reasoning generalizes from observed cases and yields conclusions with varying degrees of support, whereas deductive reasoning establishes conclusions that necessarily follow from stated premises. Mathematical induction is a deductive proof technique, despite its name. It requires a base case and a step showing $P(k) \rightarrow P(k+1).$

Theory and proof principles

A finite list of examples never proves a universal statement over an infinite domain. A single counterexample refutes it. Strong induction assumes $P(1), \ldots, P(k)$ to prove $P(k+1)$, and is logically equivalent in strength to ordinary induction for natural-number propositions.

Mathematical induction: a full proof

Claim: for every positive integer $n, 1 + 3 + \cdots + (2n-1) = n^2.$ Base case $n = 1: 1 = 1^2$. Inductive hypothesis: assume the sum of the first k odd positive integers is $k^2.$ Then the sum through the $(k+1)$st odd integer is $k^2 + (2k+1) = (k+1)^2.$ Thus the claim holds for every positive integer n. This numerical proof supplements, rather than replaces, the multi-proposition logic exercises below.

Advanced worked proofs

Proof 6.1: Five-atom deductive synthesis

Premises: $p \rightarrow q;\ q \rightarrow r;\ (r \wedge s) \rightarrow t;\ p \wedge s$

Prove: t

Proof:

1. From $p \wedge s$ extract p and s.

2. From $p \rightarrow q$ and p infer q.

3. From $q \rightarrow r$ and q infer r.

4. Conjoin r and s.

5. Apply $(r \wedge s) \rightarrow t$ to conclude t.

Therefore the conclusion follows from the premises.

Chapter review

The examples demonstrate that complex formulas should be decomposed into smaller statements and that each inference must be justified. The distinction between validity, truth, and equivalence remains essential.

Example

Check using the logical equation that

$$[(p \vee q \vee \sim r) \wedge (\sim p \vee q \vee r) \wedge (p \wedge \sim q \vee r)] \rightarrow (p \wedge q)$$

Prove $$\mathrm{A} = (\mathrm{p} \vee \mathrm{q} \vee \sim \mathrm{r})$$

$$\mathrm{B} = (\sim \mathrm{p} \vee \mathrm{q} \vee \mathrm{r})$$

$$\mathrm{C} = (\mathrm{p} \vee \sim \mathrm{q} \vee \mathrm{r})$$

Therefore, the specified proposition can be abbreviated as

$$(\mathrm{a} \wedge \mathrm{b} \wedge \mathrm{c}) \rightarrow (\mathrm{p} \wedge \mathrm{q})$$

$$\equiv \sim (a \wedge b \wedge c) \vee (p \wedge q)$$

Instead of value A,B,C

$$\equiv \sim [(p \vee q \vee \sim r) \wedge (\sim p \vee q \vee r) \wedge (p \vee \sim q \vee r)] \vee (p \wedge q)$$

$$\equiv \sim (p \vee q \vee \sim r)$$

$$\vee \sim (\sim p \vee q \vee r)$$

$$\vee \sim (p \vee \sim q \vee r)$$

$$\vee (p \wedge q)$$

Instead of value

$$[(\mathrm{p} \vee \mathrm{q} \vee \sim \mathrm{r}) \wedge (\sim \mathrm{p} \vee \mathrm{q} \vee \mathrm{r}) \wedge (\mathrm{p} \wedge \sim \mathrm{q} \vee \mathrm{r})] \rightarrow (\mathrm{p} \wedge \mathrm{q})$$

$$\equiv (\sim p \wedge \sim q \wedge r) \vee (p \wedge \sim q \wedge \sim r) \vee (\sim p \wedge q \wedge \sim r) \vee (p \wedge q)$$

Instead of the value $p = T, q = F, r = T$ in the price

$$[(P \vee Q \vee \sim R) \wedge (\sim P \vee Q \vee R) \wedge (P \vee \sim Q \vee R)] \rightarrow (P \wedge Q)$$
$$\equiv [(T \vee F \vee \sim T) \wedge (\sim T \vee F \vee T) \wedge (T \vee \sim F \vee T)] \rightarrow (T \wedge F)$$
$$\equiv [(T \vee F \vee F) \wedge (F \vee F \vee T) \wedge (T \vee T \vee T)] \rightarrow F$$
$$\equiv T \wedge T \wedge T) \rightarrow F$$
$$\equiv T \rightarrow F$$
$$\equiv F$$

As a result

$$[(P \vee Q \vee \sim R) \wedge (\sim P \vee Q \vee R) \wedge (P \vee \sim Q \vee R)] \rightarrow (P \wedge Q)$$

It's not eternal.

Example 1 : Prove that

$$\frac{6}{(n^3 - n)}$$

for every positive integer n

Proof

When $n = 1$

$$1^3 - 1 = 0$$

Which is divisile by 6

Assume

$$\frac{6}{(k^3 - k)}$$

$$(k+1)^3 - (k+1) = k^3 + 3k^2 + 3k + 1 - k - 1$$

$$= (k^3 - k) + 3k(k+1)$$

The first term is divisible by 6 by the induction hypothesis

Sinc one of k and $k + 1$ is even

$k(k+1)$ Is even

Hance $3k(k+1)$ is divisible by 6

Therefore $\frac{6}{((k+1)^3 - (k+1))}$

Thus $\frac{6}{(n^3 - n)}$

Imductive

Example 2

Prove by mathematical induction that for all inteers $n \geq 1$:

$$1^3 + 2^3 + 3^3 + \cdots + n^3 = \left[\frac{n(n+1)}{2}\right]^2$$

Proof Inductive Hypothesis $(n = k)$:

$$1^3 + 2^3 + 3^3 + \cdots + K^3 = \left[\frac{k(k+1)}{2}\right]^2$$

Inductive Step $(n = k + 1)$:

$$[1^3 + 2^3 + \cdots + k^3] + (k+1)^3$$

$$= \left[\frac{k(k+1)}{2}\right]^2 + (k+1)^3$$

$$= \frac{k^2(k+1)^2}{4} + (k+1)^3$$

$$= (k+1)^2\left[\frac{k^2}{4} + (k+1)\right]$$

$$= (k+1)^2\left[\frac{k^2 + 4k + 4}{4}\right]$$

$$= (k+1)^2\left[\frac{(k+2)^2}{4}\right]$$

$$= \left[\frac{(k+1)(k+2)}{2}\right]^2$$

Conclusion: Prove that the equation is true for the power that $k + 1$

In conclusion, this equation is true for all counts. $N \geq 1$ completely

Deductive

Example 3

Is the following argument valid ?

- **Premise 1:** $p \rightarrow (q \vee r)$
- Premise 2: $q \rightarrow \sim s$
- Premise 3: s
- Conclusion : $\therefore p \rightarrow r$

Proof Assume that this justification is unreasonable by determining that all reasons are true (T) and the conclusion is false (F) :

$$\left[[p \rightarrow (q \wedge r)] \wedge (q \rightarrow \sim s) \wedge s\right] \rightarrow (p \rightarrow r) \equiv F$$

Because the main connector is $\rightarrow$ and the result is F, there is only one case that all the reasons are true (T) and the conclusion is false (F) :

$$[p \rightarrow (q \vee r)] \wedge (q \rightarrow \sim s) \wedge s \equiv T \text{ and } p \rightarrow r \equiv F$$

Consider the false conclusion $(P \rightarrow r = F)$: Bi clause "**If... Then**" can be false. The only case is that the front is true and the latter is False :

$$p \equiv T \; and \; r \equiv F$$

Consider the true cause, which is connected with "and" $(\wedge)$, making every cause must have a true value at the same time:

$$p \rightarrow (q \vee r) \equiv T, g \rightarrow \sim s \equiv T, and \; s \equiv T$$

Substitute the value $s \equiv T$ in the second reason $(q \rightarrow \sim s) \equiv T$:

$$q \rightarrow \sim T \equiv T \Longrightarrow q \rightarrow F \equiv T$$

The solution if q then F can be true. Force the face (q) to be false only:

$$q \equiv F$$

Replace the value $p \equiv T, q \equiv F, r \equiv F$ in the first reason $[P \rightarrow (q \vee r)] \equiv T$:

$$T \rightarrow (F \vee F) \equiv T$$

$$T \rightarrow F \equiv T$$

But compared to the logical rule of $\mathrm{T} \rightarrow F$, which results in false (F), there is a contradiction $(F \equiv T)$. All cases of assumptions that make unreasonable reasoning lead to contradictions. It is a valid argument.

References

Epp, S. S. (2011). Discrete Mathematics with Applications (4th ed.). Brooks/Cole.

Rosen, K. H. (2019). Discrete Mathematics and Its Applications (8th ed.). McGraw-Hill Education.

Enderton, H. B. (2001). A Mathematical Introduction to Logic (2nd ed.). Academic Press.

Velleman, D. J. (2019). How to Prove It: A Structured Approach (3rd ed.). Cambridge University Press.